

Ê PHASOR TEMPLATES (Air: ε_r=μ_r=1)

Wavenumber per medium *nonmag.*

$$k_i = \frac{\omega\sqrt{\epsilon_{r,i}}}{c} \text{ (rad/m)} \Rightarrow k_2 = k_1 \sqrt{\epsilon_r/2/\epsilon_{r,1}}$$

Direction table *u* ∈ {x, y, z} = axis perp. to boundary

Inc dir	Inc	Refl	Trans
+û	$e^{-jk_1 u}$	$e^{+jk_1 u}$	$e^{-jk_2 u}$
-û	$e^{+jk_1 u}$	$e^{-jk_1 u}$	$e^{+jk_2 u}$

+û direction ⇔ Med 2 at u ≥ 0. Reflected sign flips, transmitted matches incident.

General template (incident has pol. ê, amplitude a₀):

Êⁱ = a₀ê e^{±jk₁u} (V/m) (incident wave, Med 1)

Ê^r = Γ a₀ê e^{±jk₁u} (V/m) (reflected wave, Med 1)

Ê^t = τ a₀ê e^{±jk₂u} (V/m) (transmitted wave, Med 2)

(lossy med 2: replace e^{±jk₂u} with e^{±α₂u} e^{±jβ₂u})

Total in medium 1: Ê₁ = Êⁱ + Ê^r.

ê by polarization *plug into templates above*

Pol.	ê	Note
Lin. x̂	x̂	E _y = 0
Lin. ŷ	ŷ	E _x = 0
Lin. at 45°	(x̂ + ŷ)/√2	in phase
RHC	x̂ - jŷ	δ = -π/2
LHC	x̂ + jŷ	δ = +π/2
General	a _x x̂ + a _y e ^{jδ} ŷ	elliptical

Γ, τ act on ê unchanged. Only swap the polarization, not the formulas.

TABLE 8-2 — Γ, τ, R, T SUMMARY

Med. 1 (z < 0) incident side; **Med. 2** (z ≥ 0) transmitted.
Shorthand: c_i = cos θ_i, c_t = cos θ_t.

	Normal (θ _i = 0)	⊥ pol	∥ pol
Γ refl. wave	$\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\frac{\eta_2 c_i - \eta_1 c_t}{\eta_2 c_i + \eta_1 c_t}$	$\frac{\eta_2 c_t - \eta_1 c_i}{\eta_2 c_t + \eta_1 c_i}$
τ trans. wave	$\frac{2\eta_2}{\eta_2 + \eta_1}$	$\frac{2\eta_2 c_i}{\eta_2 c_i + \eta_1 c_t}$	$\frac{2\eta_2 c_t}{\eta_2 c_t + \eta_1 c_i}$
τ rel.	τ = 1 + Γ	τ _⊥ = 1 + Γ _⊥	τ _∥ = (1 + Γ _∥) $\frac{c_i}{c_t}$
R refl. pwr	Γ ²	Γ _⊥ ²	Γ _∥ ²
T trans. pwr	τ ² $\frac{\eta_1}{\eta_2}$	τ _⊥ ² $\frac{\eta_1 c_t}{\eta_2 c_i}$	τ _∥ ² $\frac{\eta_1 c_t}{\eta_2 c_i}$
R + T total	T = 1 - R	T _⊥ = 1 - R _⊥	T _∥ = 1 - R _∥

Notes: sin θ_t = √μ₁ε₁/μ₂ε₂ sin θ_i; *nonmag.* ⇒ η₂/η₁ = n₁/n₂.

Intrinsic impedance η_i *plug into table above*

$$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \text{ (}\Omega\text{)} \quad \eta_0 = 120\pi \approx 377 \Omega$$

Default μ_r = 1 (air, dielectric). Use μ_r ≠ 1 only if problem says “ferromagnetic”/iron/ferrite/etc.

Low-loss correction (η_c σ/(ωε) ≪ 1)

$$\eta_c \approx \sqrt{\frac{\mu}{\epsilon}} \left(1 + j \frac{\sigma}{2\omega\epsilon'}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}$$

For Γ/τ just use η₀/√ε_r. The j term is for |η_c|, θ_η. See LOSSY MEDIA — 5 CASES for all regimes.

POWER REFLECTION & TRANSMISSION

% reflected *always*

$$\%P_r = 100 |\Gamma|^2$$

% transmitted *η-ratio matters!*

$$\%P_t = 100 |\tau|^2 \frac{\eta_1}{\eta_2}$$

Check: %P_r + %P_t = 100.

If medium 2 is denser (η₂ < η₁), |τ| > 1 but %P_t < 100% because of the η₁/η₂ factor.

BOUNDARY

η₁, η₂ — imp. (Ω)
Γ — refl. coeff (unitless)
τ = 1 + Γ — trans. coeff
k_i = ω√ε_{r,i}/c (rad/m)
α₂, β₂ — if med 2 lossy
z = 0 — boundary plane
η₀ = 120π ≈ 377 Ω

SNELL / OBLIQUE

θ_i, θ_t — angles from normal
n_i = √ε_{r,i} — index of refr.
t_i — slab thickness (m)
d — lateral disp. (m)
plane of inc. = plane of k̂+normal
⊥: Ê ⊥ this plane
∥: Ê ∥ this plane

WAVEGUIDE

a, b — wide, narrow dim. (m);
a > b
m, n — mode indices (int.)
f_{mn} — cutoff (Hz)
β — prop. const (rad/m)
Z_{TE}/Z_{TM} — wave imp. (Ω)
E_x, H_y — field comps (V/m, A/m)
η = η₀/√ε_r

VELOCITIES

u_{p0} = c/√ε_r — bulk phase vel.
u_p = u_{p0}/√1 - (f_c/f)²
u_g = u_{p0}√1 - (f_c/f)²
u_p u_g = u_{p0}²
t = L/u_g — travel time
c = 3 × 10⁸ m/s

POWER & UNITS

%P_r = 100|Γ|²
%P_t = 100|τ|² η₁/η₂
%P_r + %P_t = 100
σ (S/m), ε (F/m), μ (H/m)
ε₀ = 8.85 × 10⁻¹² F/m
μ₀ = 4π × 10⁻⁷ H/m
1/(36π) × 10⁻⁹ ≈ ε₀

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary *angles from normal*

$$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$$

Multilayer chain *stack of slabs* 1 → 2 → 3 → 4

$$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$$

Air-slabs-air shortcut *outer media identical*

θ_{exit} = θ_{incident}

Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).

LATERAL DISPLACEMENT

Beam offset through N slabs thickness t_i, refracted angle inside slab i

Slab-indexed θ_i = angle inside slab i

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

Angle-indexed (HW3 P3 style) θ₁ = incident, θ_{i+1} = inside slab i

$$d = \sum_{i=1}^N t_i \tan \theta_{i+1}$$

Same physics, different numbering. In HW3 P3: slab 1 contributes t₁ tan θ₂, slab 2 contributes t₂ tan θ₃. The angle in the formula is always the one the ray makes inside that slab.

Air-slab-air: exit angle returns to θ₁ but is laterally shifted by d.

LOSSY MEDIA — 5 CASES

Loss tangent ε' = ε_rε₀; ε₀ = 8.854 × 10⁻¹² F/m

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$$

Type	σ/ωε	η _c , α, β
Lossless (σ = 0)	0	η = η ₀ /√ε _r exact α = 0, β = ω√ε _r /c
Low-loss	< 10 ⁻²	η _c ≈ $\frac{\eta_0}{\sqrt{\epsilon_r}} (1 + j \frac{\sigma}{2\omega\epsilon'})$ α ≈ $\frac{\sigma\eta_0}{2\sqrt{\epsilon_r}}$, β ≈ $\frac{\omega\sqrt{\epsilon_r}}{c}$
Quasi-cond.	10 ⁻² –10 ²	exact: (η/√X) e ^{jθ} see below
Good cond.	> 10 ²	η _c ≈ (1+j)√π f μ/σ α ≈ β ≈ √π f μσ
Magnetic	any	keep full √μ _r /ε _r η ₀

For Γ/τ: drop j correction; use η₀/√ε_r. Magnetic: keep μ_r.

Low-loss quick params σ/(ωε) ≪ 1, μ_r = 1

$$\alpha \approx \frac{\sigma\eta_0}{2\sqrt{\epsilon_r}} \text{ (Np/m)}, \quad \beta \approx \frac{\omega\sqrt{\epsilon_r}}{c} \text{ (rad/m)}, \quad \eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}} \text{ (}\Omega\text{)}$$

Exact (quasi-cond) X = 1 + (ε''/ε')²

$$\alpha = \omega\sqrt{\frac{\mu\epsilon'}{2}}\sqrt{X-1}, \quad \beta = \omega\sqrt{\frac{\mu\epsilon'}{2}}\sqrt{X+1}$$
$$\theta_\eta = \frac{1}{2} \arctan \frac{\sigma}{\omega\epsilon'}$$

RECTANGULAR WAVEGUIDE — MODES

TE mode E_z = 0, H_z ≠ 0
TM mode H_z = 0, E_z ≠ 0
Dimensions a × b with a > b along x̂, ŷ; wave in ẑ.
E_x form (TE) *read m, n from arguments*

$$E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$$

coef. on x = mπ/a; coef. on y = nπ/b.

Identify TE_{m,n} vs TM_{m,n} *cos/sin pattern of E_x is identical for both!*

1. Read the problem statement (e.g. “a TE wave” → TE).
2. If m = 0 or n = 0 in the mode label ⇒ must be TE (TM forbids 0).
3. Source field H_z(x, y) given ⇒ TE. Source field E_z(x, y) given ⇒ TM.

TE allows at least one of m, n ≥ 1 (other can be 0). TM needs both m, n ≥ 1. That's why TE₁₀ exists but TM₁₀ doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

Shorthand: C_X = cos $\frac{m\pi x}{a}$, S_X = sin $\frac{m\pi x}{a}$, C_Y = cos $\frac{n\pi y}{b}$, S_Y = sin $\frac{n\pi y}{b}$. All fields carry an e^{-jβz} factor (omitted below). k_c² = (mπ/a)² + (nπ/b)².

TE mode generator: Ê_z

$$\tilde{H}_z = H_0 C_X C_Y$$
$$\tilde{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 C_X S_Y$$
$$\tilde{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 S_X C_Y$$
$$\tilde{E}_z = 0, \quad \tilde{H}_x = -\tilde{E}_y/Z_{TE}, \quad \tilde{H}_y = \tilde{E}_x/Z_{TE}$$

TM mode generator: Ê_z

$$\tilde{E}_z = E_0 S_X S_Y$$
$$\tilde{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 C_X S_Y$$
$$\tilde{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 S_X C_Y$$
$$\tilde{H}_z = 0, \quad \tilde{H}_x = -\tilde{E}_y/Z_{TM}, \quad \tilde{H}_y = \tilde{E}_x/Z_{TM}$$

TEM (plane wave) for reference: Ê_x = E₀e^{-jβz}, Ê_y = Ê_x/η, f_c N/A, k = ω√μ̵.

CUTOFF FREQUENCY

f_{mn} *m, n integers ≥ 0 (not both zero)*

$$f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \text{ (Hz)}$$

Common cutoffs (a > b)

Quantity	Formula
u _{p0}	c/√ε _r (hollow: u _{p0} = c)
TE ₁₀	u _{p0} /(2a) (dominant)
TE ₀₁	u _{p0} /(2b)
TE ₂₀	u _{p0} /a = 2f ₁₀
TE ₁₁ , TM ₁₁	$\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$

Propagation requires f > f_{mn}.

ε_r FROM DISPERSION

Given ω, β, mode (m, n) *result is unitless*

$$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right] \text{ (unitless)}$$

Inputs: ω (rad/s), β (rad/m), a, b (m), c = 3 × 10⁸ (m/s), m, n (unitless).

WAVE IMPEDANCE & H_y

$$Z_{TE} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}} \text{ (}\Omega\text{)} \quad Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$$
$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \quad H_y = \frac{E_x}{Z_{TE}} \text{ (TE mode)}$$

Z_{TE} > η always (denominator < 1); Z_{TM} < η.

GROUP VELOCITY & TRAVEL TIME

$$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2} \text{ (m/s)}$$
$$u_p = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}} \text{ (m/s)} \Rightarrow u_p u_g = u_{p0}^2 \text{ (m}^2\text{/s}^2\text{)}$$
$$t = L/u_g \text{ (s) travel time through guide of length } L$$

Higher modes have lower u_g ⇒ they arrive later. Group delay spread = sum over excited modes.